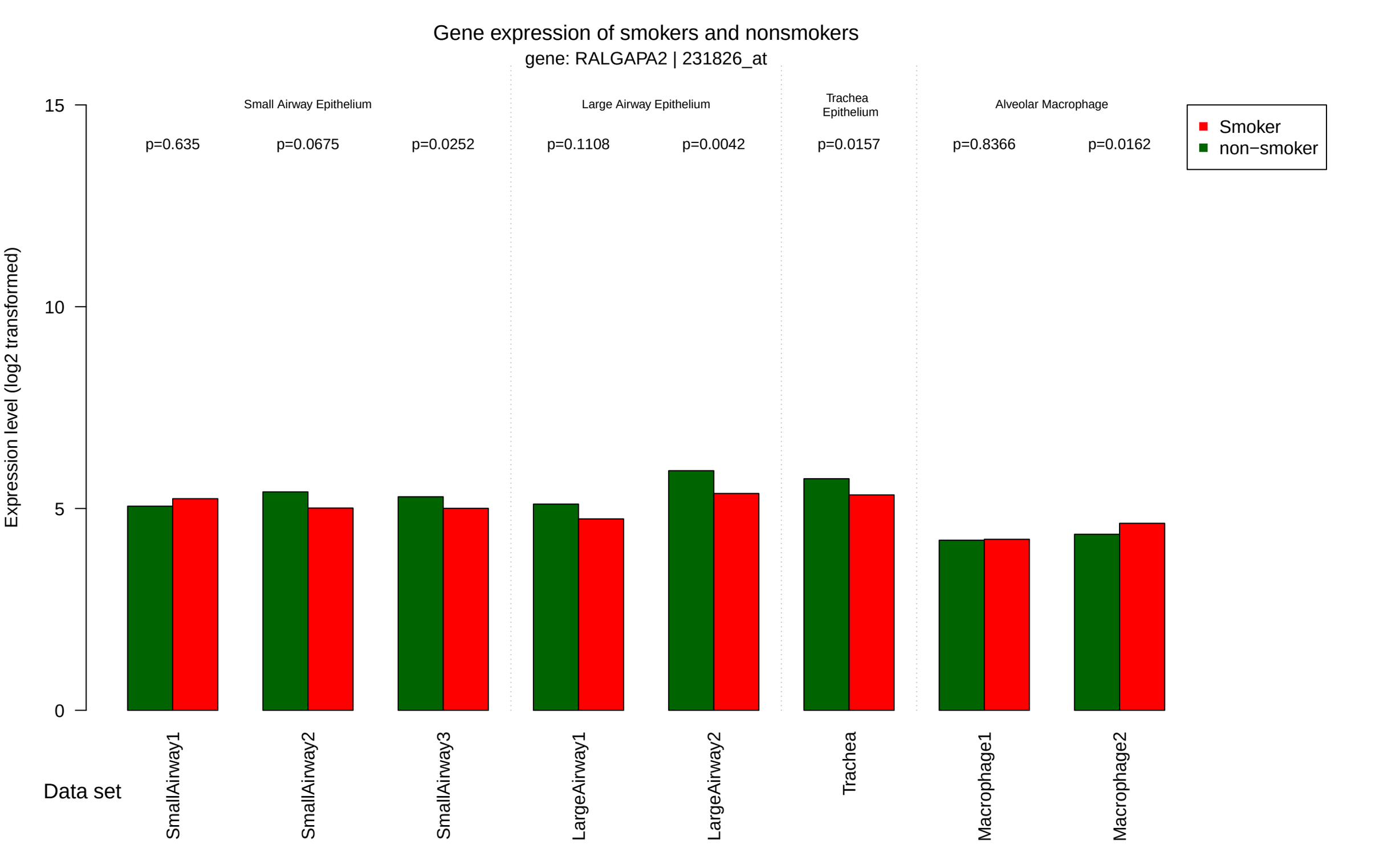
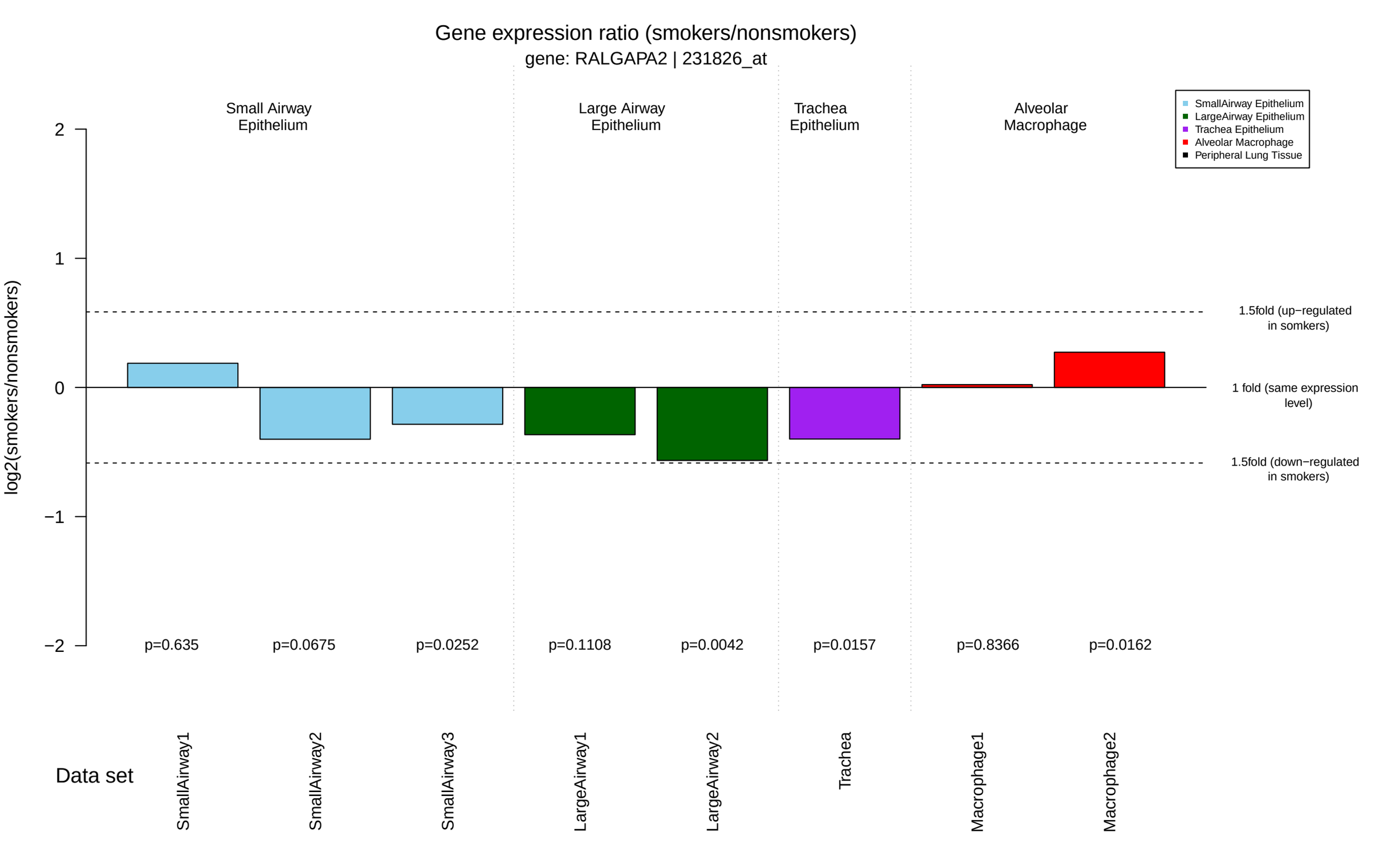
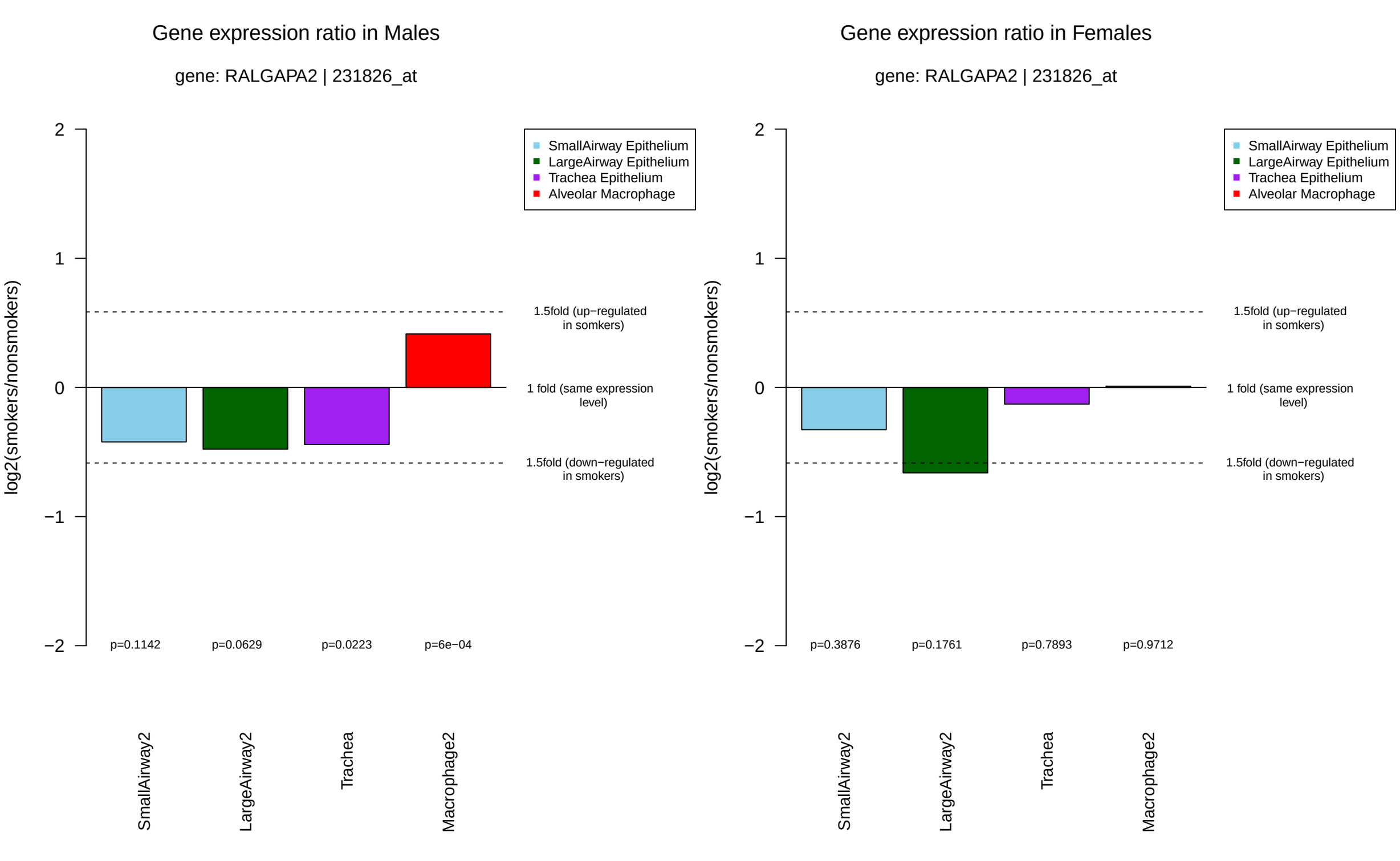
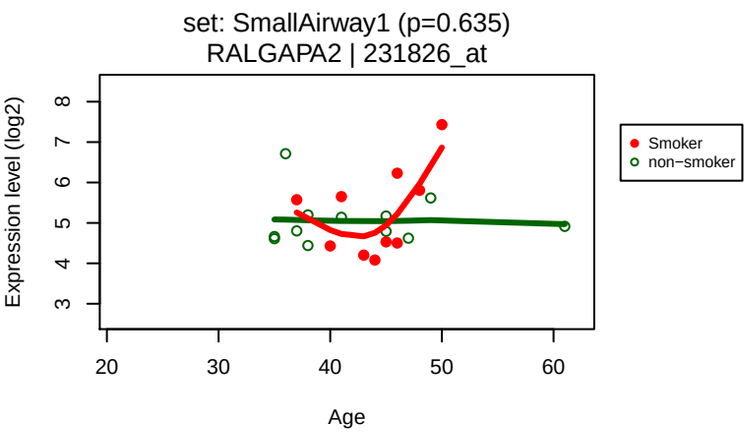
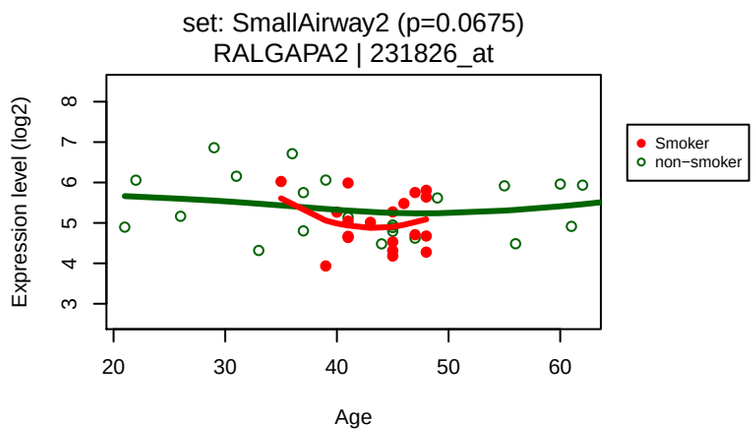
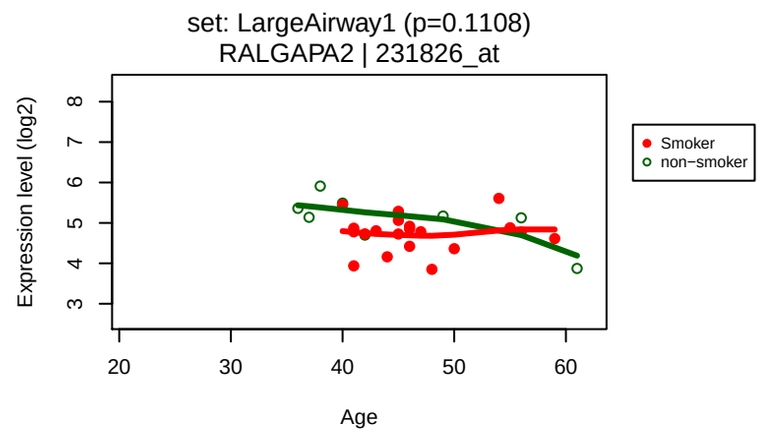

Gene expression of smokers and nonsmokers
gene: RALGAPA2 | 231826_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.635
p=0.0675
p=0.0252
p=0.1108
p=0.0042
p=0.0157
p=0.8366
p=0.0162
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

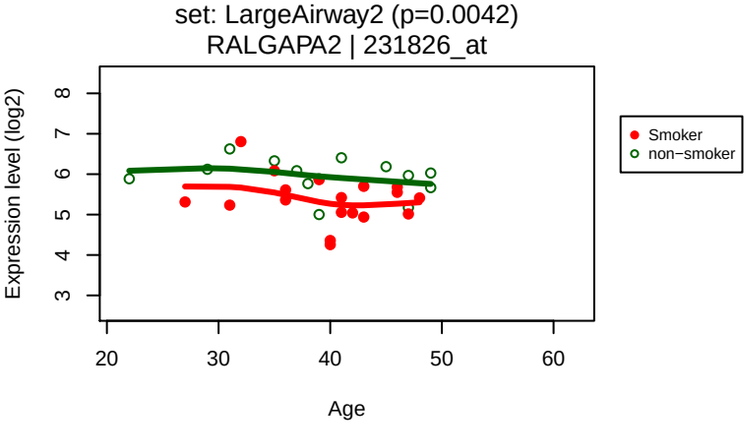

Gene expression ratio (smokers/nonsmokers)
gene: RALGAPA2 | 231826_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.635
p=0.0675
p=0.0252
p=0.1108
p=0.0042
p=0.0157
p=0.8366
p=0.0162
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

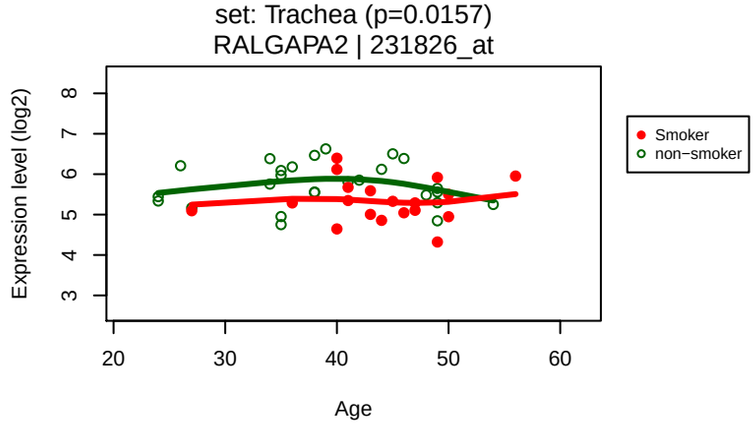

Gene expression ratio in Males
gene: RALGAPA2 | 231826_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1142
p=0.0629
p=0.0223
p=6e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGAPA2 | 231826_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3876
p=0.1761
p=0.7893
p=0.9712
SmallAirway2
LargeAirway2
Trachea
Macrophage2

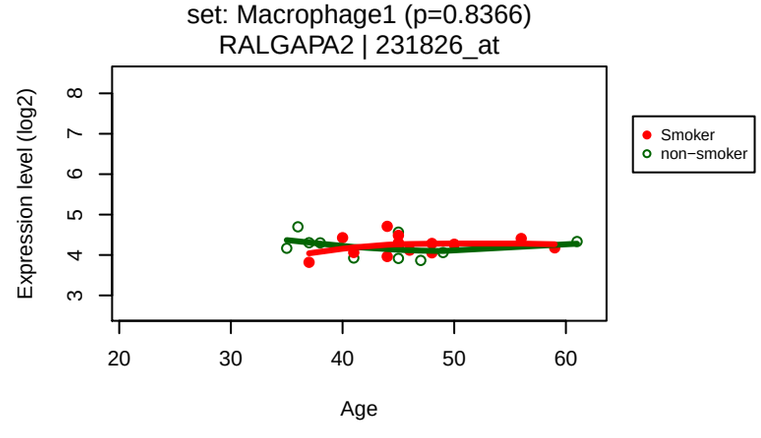

set: SmallAirway1 (p=0.635)
RALGAPA2 | 231826_at
Expression level (log2)
Age
Smoker
non−smoker

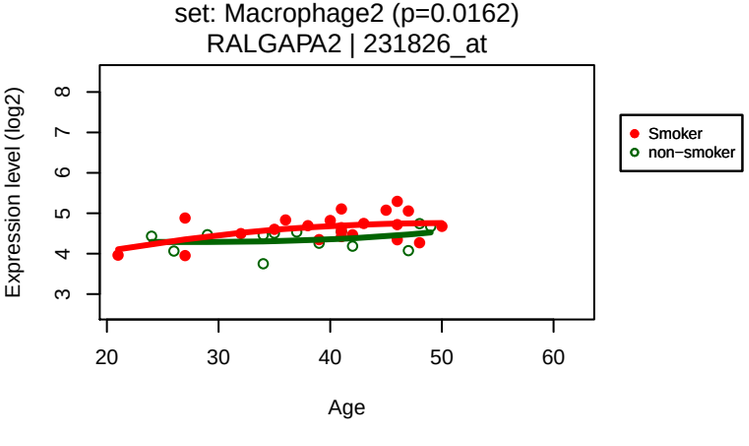

set: SmallAirway2 (p=0.0675)
RALGAPA2 | 231826_at
Expression level (log2)
Age
Smoker
non−smoker

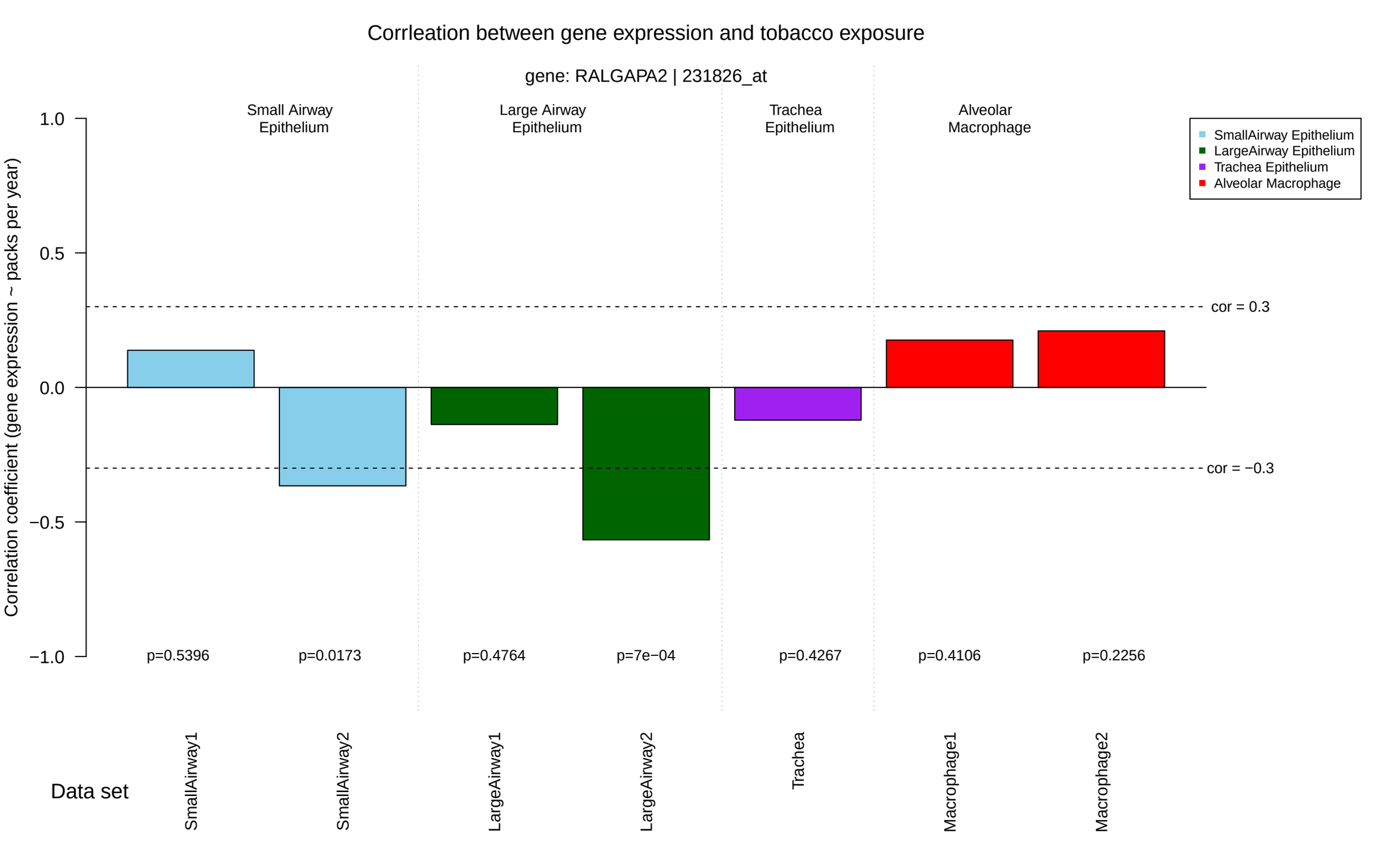

set: LargeAirway1 (p=0.1108)
RALGAPA2 | 231826_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0042)
RALGAPA2 | 231826_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0157)
RALGAPA2 | 231826_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8366)
RALGAPA2 | 231826_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0162)
RALGAPA2 | 231826_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGAPA2 | 231826_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5396
p=0.0173
p=0.4764
p=7e−04
p=0.4267
p=0.4106
p=0.2256
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2